



School of
Information Technology,
Engineering, Mathematics,
and Physics

66.2 67

CS219: Cloud Computing

Mid Semester Test Semester 2, 2024

Mode: Face to Face

Duration of Exam: 1 Hour 10 minutes

Reading Time: Extra 2 minutes

Writing Time: 1 Hour 10 minutes

Instructions:

There are 14 Short Answer Questions. Weighting: 15% | Total Marks: 70

This is a closed book exam.

Learning Outcomes

- I. Explain the capabilities and limitations of cloud computing.
- II. Choose a strategy to ensure appropriate levels of security and privacy in a cloud-based system.
- III. Explain the role of data center in cloud computing.

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public cloud → users use a shared cloud on a pay per use basis
private cloud → self hosted cloud on a company's data center for max control
hybrid cloud → mix of public and private. may host consumer apps on public and internal systems on private cloud.

Question 1

3 Marks

Explain three features of cloud computing and the cloud service deployment options for a business environment?

Three features include virtualization, which is technology that allows multiple virtual instances of IT resources to be created on a physical instance. Another feature is elasticity, which allows consumers to easily scale in-out (up-down) based on demand and the last is resiliency. Cloud services tend to have redundant services which help increase the uptime and provide high availability.

Question 2

4 Marks

Analyse the following scenario for TechCorp and recommend which service delivery model is best suited to cater for the specific needs of the organisation. Justify your answer.

- A. TechCorp is about to launch a new product line that requires additional servers, storage, and network resources. Building out this capacity in their on-premise data center would require purchasing expensive hardware and hiring additional staff for maintenance.

IaaS - instead of self-hosting everything, the company can use something like AWS to build their product. This would be suitable as they will not have to buy or maintain their hardware - AWS will handle the underlying infrastructure. This will save money & time.

- B. TechCorp's employees use various software applications (such as email, CRM, project management, and developer tools) hosted on local servers. Managing these licenses, performing updates, and ensuring seamless collaboration between teams has become a time-consuming and resource-heavy task for the IT department.

SaaS - Instead of self-hosting such software, the company can use pre-built end-to-end software and just pay on a subscription basis. Gmail could be an alternative to email and 3mg can be used as the project management software. Updates will be handled by the vendor.

Question 3

2 Marks

Evaluate the purpose of the load balancer mechanism and provide three common load balancing functions within a cloud architecture.

The load balancer mechanism is a runtime agent that distributes tasks ~~horizont~~ to different servers and is used as horizontal scaling. This increases the capacity and performance of each server. 3 common functions are workload prioritization, content-aware distribution and asymmetric distribution.

Question 4

2 Marks

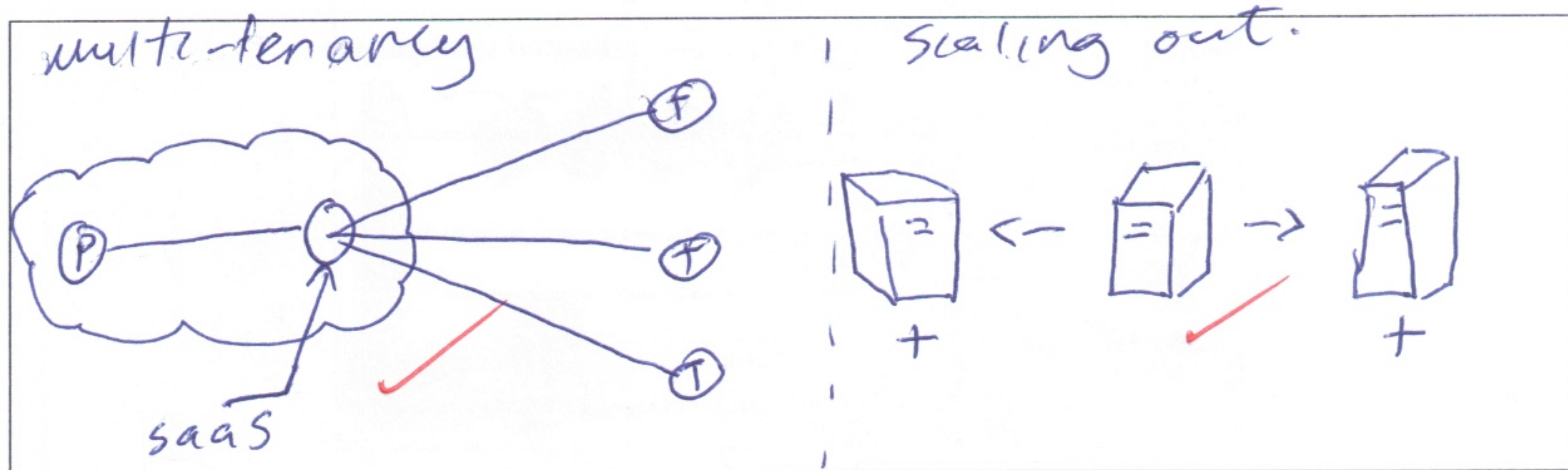
Describe the pay-per-use monitor mechanism and explain how it relates to cloud provider billing.

The pay-per-use mechanism is deployed as a monitoring or resource usage agent and collects usage metrics and sends it to a payment processing system. The payment system then uses this data and calculates billing information that is displayed on a cloud dashboard. pay per use metrics include number of VM instances created, ingress, egress, CPU usage, etc.

Question 5

P = cloud provider, T = tenant 3 Marks

Distinguish between multi-tenancy technology and scaling out resources? Provide an example of each. Use diagrams where possible.



Multi-tenancy technology allows one resource (as software in this case) to be shared with multiple consumers. Consumers have their own customization. Scaling out refers to add more IT resources of the same type to increase capacity + performance. In the diagram above, more servers are being added.

Question 6 A. Identify and explain the three APIs provided by the remote administration system to provide cloud consumers with a variety of administrative controls? 3 Marks

RAS connects to APIs exposed by underlying systems for detailed metrics.

- Billing → shows payment details based on cycles.
- Resource → shows resource usage and estimation.
- Access → allows consumers to manage and instantiate more IT based resources.

B. Describe the role of KVMs in a Unix-based operating system and how is this different from a HyperV? 2 Marks

Kernel based VMs are software that is installed onto all Linux machines and allows them to be converted to a host OS so they can create guest OS. It is different from Hyper-V as KVMs are pre-installed but Hyper-V has to be installed manually (either software/hardware based).

C. Define AAA in the context of cloud computing security? Evaluate two authentication methods used in AAA cloud security architecture. 3 Marks

- The AAA security architecture is a common way cloud environments are secured. It consists of:
 - Authentication → verifying a user's identity.
 - Authorization → checking what roles and access user has.
 - Accounting → keeping track of user activities as logs.

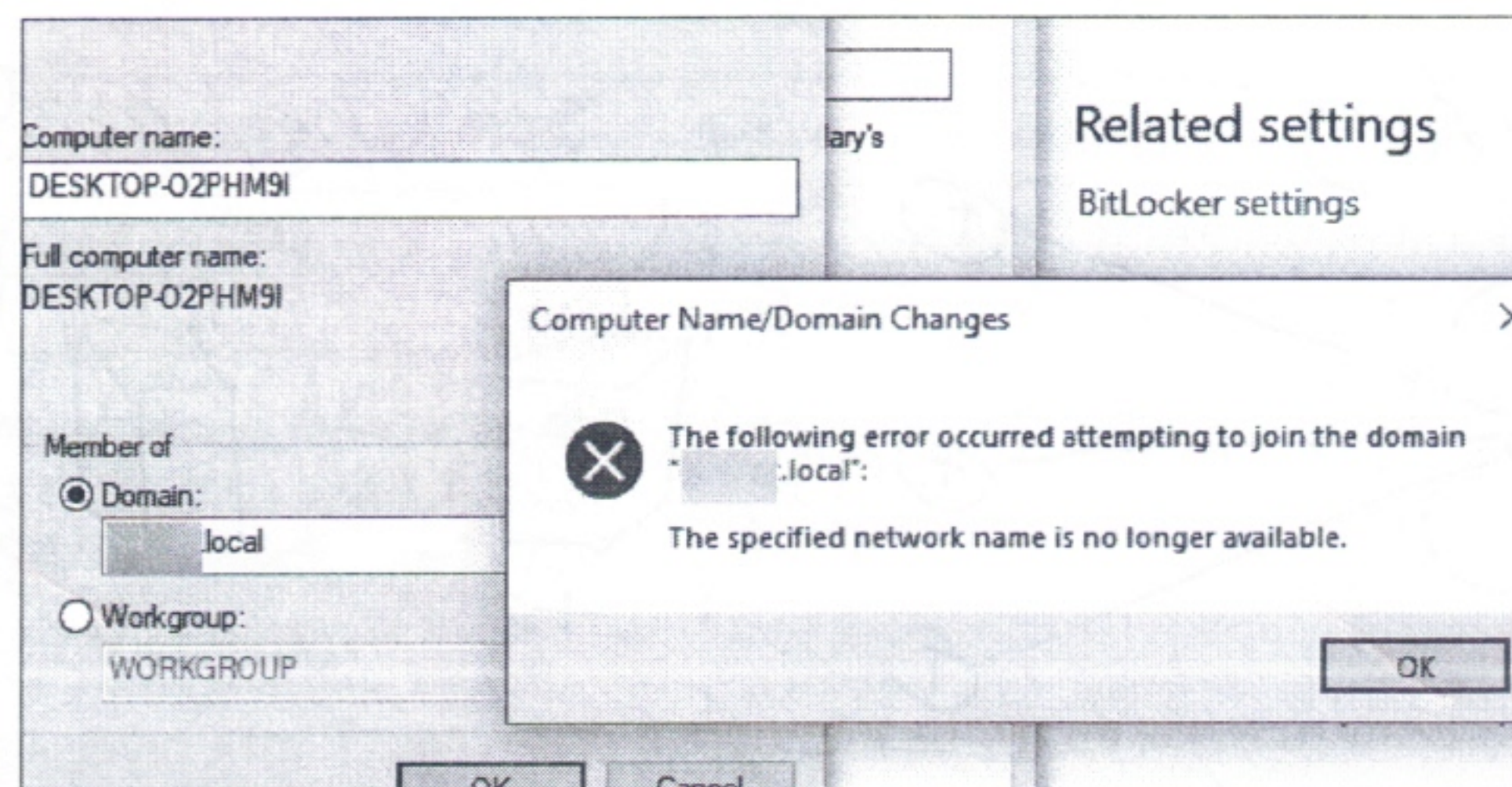
- 2 common authentication methods used in AAA are:

- PAP - password authentication protocol - This is where a user's password is sent as plain ASCII text and is not encrypted.

CHAP - challenge handshake authentication protocol - this is a more secure way of sending passwords where the server sends the client a challenge (number) and the client hashes the user password and the challenge. The server then checks if the hash matches.

Question 7

Use the diagram below to answer the following questions:



A. Provide three reasons that might have resulted in the above error.

3 Marks

- 1) wrong adapter - a host only adapter is required for VM to VM communication, which may not be enabled.
- 2) incorrect IP config - The DNS server of the client must point to the IP address of the server which may not be configured.
- 3) the wrong domain name might have been entered

B. Recommend two solutions that would be appropriate to fix the error shown in the diagram.

2 Marks

- 1) ensure host only and bridged adapters are enabled.
- 2) ensure WAN is untouched and only LAN IP is changed.
- 3) ensure DNS of server points to IP address of client

Question 8

2 Marks

Identify the infrastructure component responsible for the following:

- i. Coordinating the server hardware so that multiple virtual server instances can be created from the most convenient underlying physical server.

Virtual Infrastructure Manager (VIM)

- ii. Limited to one physical server and can therefore only create virtual images of that server.

Hypervisor (VMM)

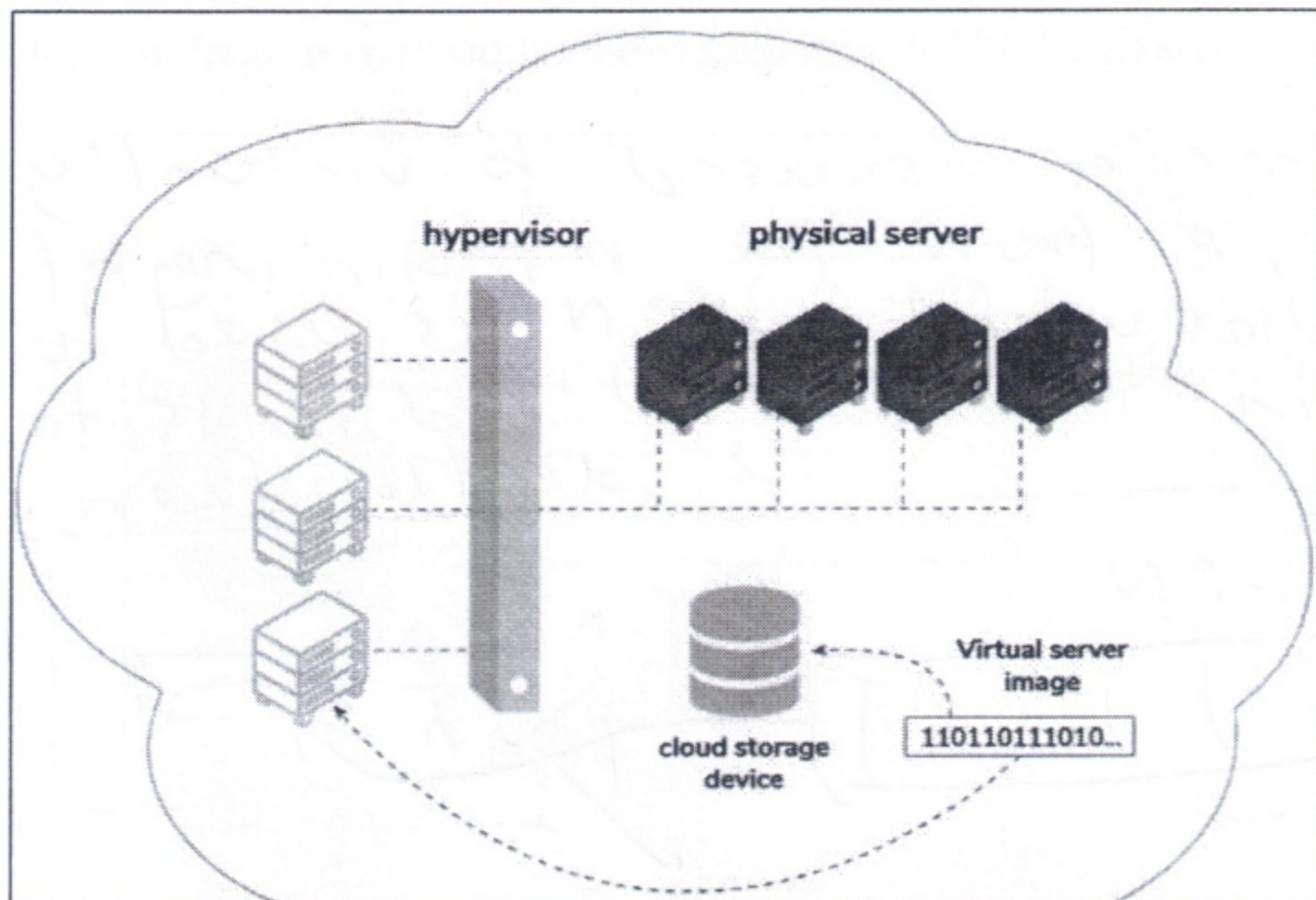
- iii. Limited virtual server management features, such as increasing the virtual server's capacity or shutting it down.

Hypervisor

- iv. Provides a range of features for administering multiple hypervisors across physical servers.

Virtual Infrastructure Manager

Question 9



A. Describe the process illustrated in the diagram above.

2 Marks

Resource replication - virtual instances of the a physical resource is being created. In this case, the hyper-v is using virtual server images stored on a cloud storage device to replicate more virtual servers.

B. Contrast the roles of the virtual infrastructure manager and virtual machine monitor? 2 Marks

A virtual machine manager is used to create and manage virtual machines. It is limited to only one type of servers which is where virtual infrastructure managers come in. VIMs are used to manage the VMs as they have more features and this allows for different types of servers.

C. Write down one limitation of a hypervisor.

1 Mark

1) A hyper-v can only create one type of server due to hardware limitations. eg Dell physical machine can only create the same Dell virtual machine

D. List two types of database storage interfaces available in the cloud?

2 Marks

Relational (SQL) and Non-relational (NoSQL).

E. Compare and contrast between the primary portals of remote administration?

2 Marks

Resource usage portal shows IT resource usage data like CPU usage and Billing portal shows payments due based on cycles.

F. Explain the information generated through the monitoring agent and polling agent of cloud service administration?

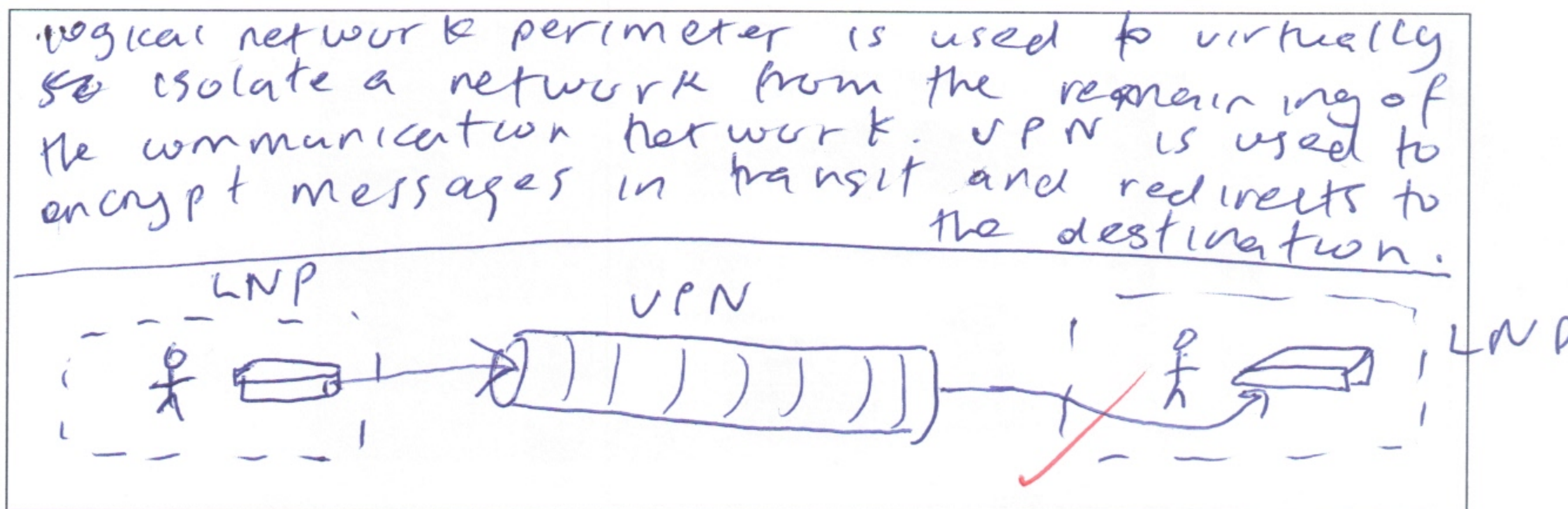
2 Marks

monitoring agent generates traffic and network data like web analytics and polling agent generates system status data like uptime and downtime.

Question 10

3 Marks

Illustrate the logical network perimeter and the role of a VPN. Use a diagram to support your answer.



Question 11

3 Marks

Classify the following the following as either Infrastructure as a Service (IaaS), Platform as a Service (PaaS) or Software as a Service (SaaS):

a. AWS EC2 -

IaaS ✓

b. Google Compute Engine -

IaaS ✓

d. Prime Video (Amazon) -

SaaS ✓

e. eBay -

SaaS ✓

f. Google App Engine -

PaaS ✓

g. Microsoft Azure VM -

IaaS ✓

Question 12

7 Marks

Consider the scenario where the cloud consumer has created an instance in Amazon Web Service (AWS). Clearly indicate whether the consumer or vendor is responsible for managing the following components of cloud offerings:

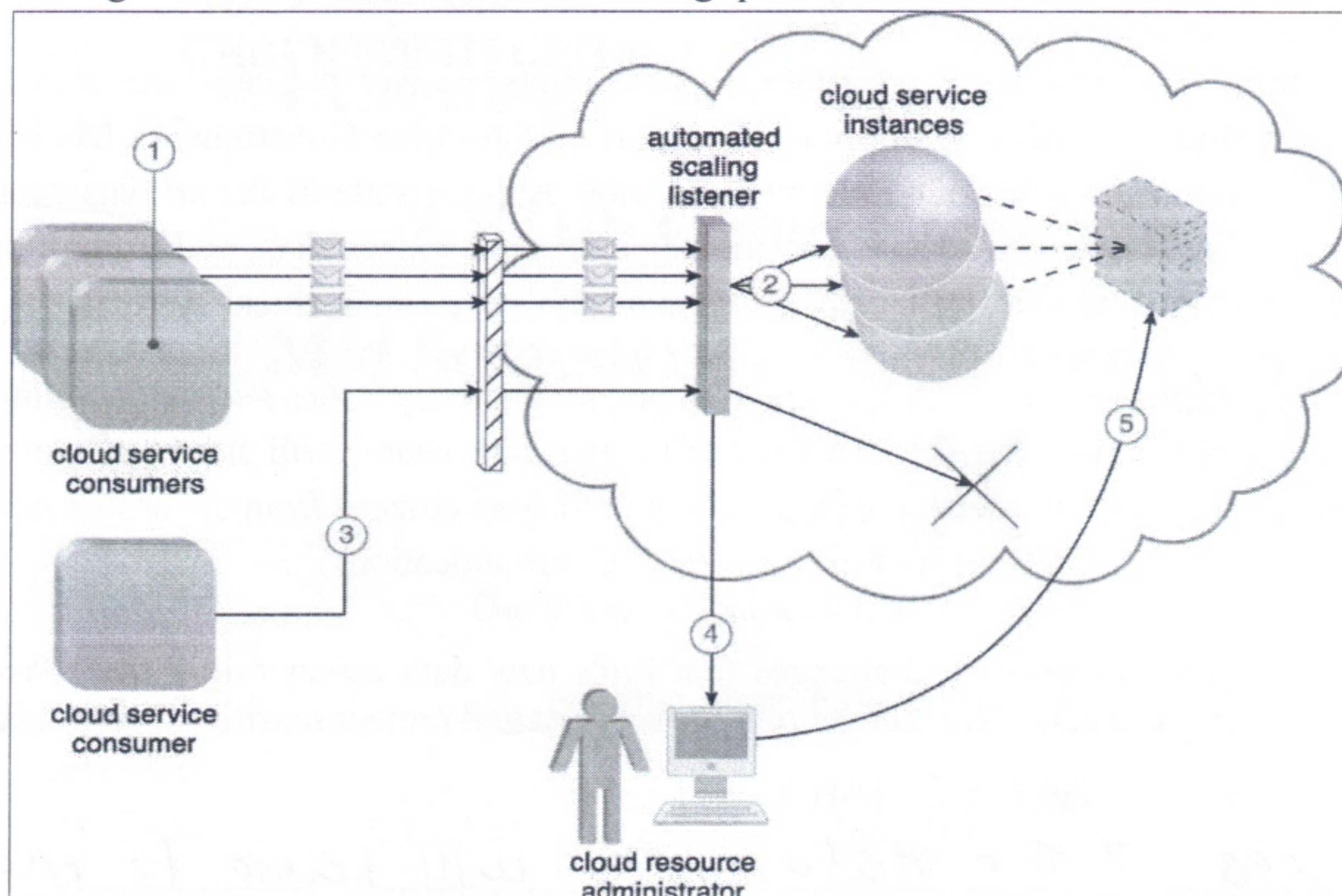
Infrastructure as a Service (IaaS)			Software as a Service (SaaS)			
Operating System	Data	Applications	Networking	Storage	Virtualisation	Servers
C ✓	C ✓	C ✓	✓	✓	✓	✓

Platform as a Service (PaaS)						
Networking	Storage	Virtualisation	Servers	Operating System	Data	Applications
✓	✓	✓	✓	✓	C	C

C = consumer
V = vendor ✓

Question 13

Use the diagram below to answer the following questions:



A. Explain the process labelled 1 to 5 in the diagram above.

2 Marks

Automatic Scaling Listener - (1) cloud consumers send requests to cloud service and ASL scales out (2) and creates 3 instances. (3) a fourth consumer tries to access the cloud service but is unable to because (4) the owner had set a limit of 3 instances via the cloud console. This unavailability is sent back to the admin which then forwards the issue to the physical server (5) after.

B. Consider a trusted attacker overloading packets to a virtual server in the diagram above. Analyse the impact it could have on other cloud consumers. Recommend countermeasures against such an attack.

3 Marks

The scenario above is a DoS attack and this will impact other cloud consumers as those that were sharing the same IT resource (multitenancy) will not be able to access that resource. Counter measures include (a) setting up failover plan with active-active strategy and (b) implementing rate limiting and DoS protection using services like Cloudflare.

C. Describe the purpose of an automated scaling listener.

2 Marks

It is a mechanism that is located near a firewall and using the network traffic data, it can either (a) automatically scale resources up or (b) alert the owner so they can manually adjust resource allocation.

Question 14

Case Study: Fiji's National Digital Transformation Initiative

Background: The government of Fiji, in its efforts to improve public service delivery, has launched a National Digital Transformation Initiative. According to a report by the Fijian Government's Ministry of Communications, a key component of this initiative involves establishing a state-of-the-art data center in Suva. This data center will host critical government applications, manage citizen data, and support public service platforms, including e-governance, online tax filing, and healthcare management systems.

Fiji's geographic location presents unique challenges, such as potential latency issues and vulnerabilities to natural disasters, which could impact the QoS. Additionally, the data center will need to implement stringent security measures to protect the sensitive personal data of Fijian citizens from cyber threats.

(Source: Fiji Ministry of Communications)

- A. Analyse the potential QoS and security challenges that Fiji's new data center might face. Propose strategies to ensure that the data center can deliver reliable services and protect sensitive data in a cloud-based environment.

5 Marks

- latency issues → the datacenter will have to make use of high performance routers and negotiate some solutions with local ISPs.
- natural disasters → the datacenter will have to make use of clustering technology to ensure failover resources are available ~~when~~ in case of a disaster.
- security → the datacenter will need strong physical security, along with a secure cyber setup with potentially ~~AAA~~ and zero trust architecture.

- B. Evaluate the relevance of a data center in cloud computing. Also include at least three technologies commonly used in a data center for providing a cloud service.

5 Marks

Data centers are a core cloud enabling technology. It is the collection of high power IT resources in a centralized location. A cloud provider like AWS might have multiple data centers around the world in various regions and in various countries.

3 common technologies commonly used are:

- a) Standardized hardware → this is the use of standard and modular computing hardware as it is easier to find repair parts / replacements.
- b) Storage hardware → data centers also require extremely large amounts of storage and with fast speeds. They make use of SANs and NASs to achieve this (this counts as networking hardware too).
- c) Security → datacenters offer various levels of security, ranging from physical security (like biometrics) to digital security (like aging passwords) to ensure data integrity and confidentiality is maintained.

END OF EXAM